H2 Math Summary

Example Questions

Maximus Pung

Contents

1 Systems of Linear Equations 2

Question 1.1 Forming equations from a table . . . 2
Question 1.2 Unknown coefficients of a polynomial . . . 3
Question 1.3 Unknown constants in a recurrence relation . . . 4
Question 1.4 More equations than variables . . . 5
Question 1.5 Fewer equations than variables with an inequality . . . 6

2 Inequalities 7

Question 2.1 Graphical method and sign test . . . 7
Question 2.2 Sign test and $\frac{1}{x}$ substitution . . . 8
Question 2.3 Modulus inequality and substitution . . . 9
Question 2.4 Modulus graphs and area . . . 10

3 Graphing Techniques 11

Question 3.1 Range of values y cannot take and rational curve sketch . . . 11
Question 3.2 Stationary point condition and line through intersection of asymptotes . . . 13
Question 3.3 Increasing rational function and superimposed hyperbola . . . 15
Question 3.4 Parametric sketch with endpoint and intersection with another curve . . . 17
Question 3.5 Cartesian equation of parametric curve and its oblique asymptotes . . . 19

4 Graphing Transformations 21

Question 4.1 Reverse transformation to find the original equation . . . 21
Question 4.2 Translation, scaling, modulus and reflection sketches . . . 22
Question 4.3 Graphs of $\mathrm{f}'(x)$, $\frac{1}{\mathrm{f}(x)}$ and $|\mathrm{f}(x)|$ from $y = \mathrm{f}(x)$ . . . 24
Question 4.4 Rational graph and $y = \frac{1}{x}$ by a sequence of transformations . . . 26

5 Functions 27

Question 5.1 Modulus linear/linear function, range and inverse . . . 27
Question 5.2 Self-inverse function, f^{2025} and range of a composite . . . 29
Question 5.3 Periodic piecewise function: sketch and area . . . 31
Question 5.4 Composite with a piecewise function . . . 32
Question 5.5 Existence, rule, domain and range of a composite function . . . 34
Question 5.6 One-one test for a function defined on integers . . . 36

6 Sequences and Series 37

Question 6.1 Standard formulas, change of index and divergence . . . 37
Question 6.2 Convergence and change of index . . . 38
Question 6.3 Arithmetic sequence from a recurrence relation, sum of even-numbered terms and a GP . . . 39

1 Systems of Linear Equations

1.1 Three different blends of coffee beans – Blend A, Blend B and Blend C – are sold at both M Café and I Café. The usual selling prices of the blends are the same at both cafés. The total cost of buying one package each of Blend A, Blend B and Blend C is \$176. During a holiday sale the two cafés offered the following discounts:

Café	Discounts given for each package			Total price after the discount
	Blend A	Blend B	Blend C	
M Café	15%	10%	5%	\$161.80
I Café	7%	5%	2%	\$169.56

(i) Find the usual selling price of each package of Blend A, Blend B and Blend C coffee beans respectively. [3]

[MI 2025 Prelim P1 Q4]

Solution		Comments
(i) **[3]**	Let the usual prices be \$$a$, \$$b$, \$$c$. $a+b+c=176 \quad ---(1)$ $0.85a+0.90b+0.95c=161.80 \quad ---(2)$ $0.93a+0.95b+0.98c=169.56 \quad ---(3)$ Using GC, $a=32,\ b=44,\ c=100$ $\therefore$ Blend A: \$32, Blend B: \$44, Blend C: \$100	

1.2 A function f is defined by $\mathrm{f}(x)=ax^3+bx^2+cx+d$ where a, b, c and d are constants. The graph of $y=\mathrm{f}(x)$ has a turning point at $x=-1$ and passes through the points $(2, 10)$ and $(-2, -2)$. Given also that $\int_{-2}^{2} \mathrm{f}(x)\,\mathrm{d}x=16$, find the values of a, b, c and d. [4]

[SAJC 2025 Prelim P1 Q1]

Solution		**Comments**
[4]	$\mathrm{f}'(x)=3ax^2+2bx+c$ $\mathrm{f}'(-1)=0$ $3a-2b+c=0 \quad ---(1)$ $\mathrm{f}(2)=10$ $8a+4b+2c+d=10 \quad ---(2)$ $\mathrm{f}(-2)=-2$ $-8a+4b-2c+d=-2 \quad ---(3)$ $\int_{-2}^{2} (ax^3+bx^2+cx+d)\,\mathrm{d}x=16$ $\left[\frac{a}{4}x^4+\frac{b}{3}x^3+\frac{c}{2}x^2+dx\right]_{-2}^{2}=16$ $(4a+\frac{8}{3}b+2c+2d)-(4a-\frac{8}{3}b+2c-2d)=16$ $\frac{16}{3}b+4d=16 \quad ---(4)$ Using GC, $a=3,\ b=0,\ c=-9,\ d=4$	

Two linear equations are independent if and only if neither equation is a linear combination of the other.

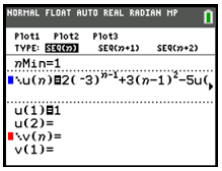
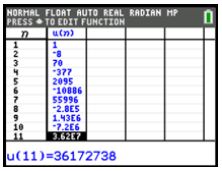
1.3 A sequence of real numbers, x_n, satisfies the recurrence relation

$$x_{n+1} = a(-3)^n + bn^2 + cx_n, \text{ for } n, a, b, c \in \mathbb{Z} \text{ and } n \geqslant 1.$$

Given that $x_1 = 1$, $x_2 = -8$, $x_3 = 70$, and $x_4 = -377$, find the values of a, b and c. [3]

Hence, find the values of x_{10} and x_{11}. [2]

[HCI 2025 Prelim P1 Q2]

Solution		**Comments**
[2]	$n = 1$ $x_2 = a(-3)^1 + b(1)^2 + cx_1$ $-8 = -3a + b + c \quad ---(1)$ $n = 2$ $x_3 = a(-3)^2 + b(2)^2 + cx_2$ $70 = 9a + 4b - 8c \quad ---(2)$ $n = 3$ $x_4 = a(-3)^3 + b(3)^2 + cx_3$ $-377 = -27a + 9b + 70c \quad ---(3)$ Using GC, $a = 2,\ b = 3,\ c = -5$ $x_{n+1} = 2(-3)^n + 3n^2 - 5x_n$ Using GC, $x_{10} = -7\,210\,868$ $x_{11} = 36\,172\,738$   	

1.4 The lines l_1 and l_2 have equations

$$l_1: \mathbf{r} = \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} -2 \\ 1 \\ 2 \end{pmatrix},$$

$$l_2: \mathbf{r} = \begin{pmatrix} 0 \\ 5 \\ -8 \end{pmatrix} + \mu \begin{pmatrix} 2 \\ -2 \\ 3 \end{pmatrix},$$

where λ and μ are parameters.

(i) Show that l_1 and l_2 are skew lines. [2]

[SAJC 2025 Prelim P1 Q3]

Solution		**Comments**
(i) **[2]**	$1-2\lambda = 2\mu$ $---(1)$ $2+\lambda = 5-2\mu$ $---(2)$ $2\lambda = -8+3\mu$ $---(3)$ Solve (1) and (2): $\lambda = -2,\ \mu = \frac{5}{2}$ Sub into (3): $\text{LHS} = 2(-2) = -4$ $\text{RHS} = -8+3\left(\frac{5}{2}\right) = -\frac{1}{2}$ $\text{LHS} \neq \text{RHS},$ no solution $\therefore l_1$ and l_2 do not intersect $\because \begin{pmatrix} -2 \\ 1 \\ 2 \end{pmatrix} \neq k \begin{pmatrix} 2 \\ -2 \\ 3 \end{pmatrix}$ for any $k \in \mathbb{R}$ $\therefore l_1$ and l_2 are not parallel $\therefore l_1$ and l_2 are skew (shown)	

1.5 A movie theatre supports a charity event by providing discounted movie tickets at the following prices based on age group.

Age group	Discounted ticket price
$\leqslant 12$ years	\$6
13 to 49 years	\$9
$\geqslant 50$ years (senior citizen)	\$4

The event organiser plans to spend \$700 to sponsor 100 participants of which at least 35 of them are senior citizens. Find the possible number of participants for each age group. [4]

[TJC 2022 Prelim P2 Q1]

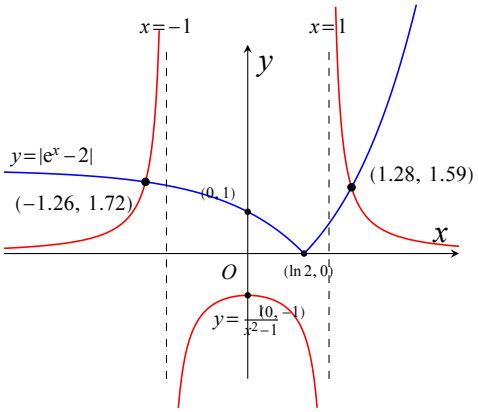
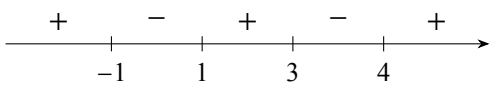
Solution		**Comments**
[4]	Let x, y, z be the number of participants aged $\leqslant 12$, 13 to 49 and $\geqslant 50$ years respectively. $\left.\begin{aligned} x+y+z&=100 \quad ---(1)\\ 6x+9y+4z&=700 \quad ---(2)\end{aligned}\right\} \Rightarrow$ Using GC, $x=\frac{200}{3}-\frac{5}{3}z,\ y=\frac{100}{3}+\frac{2}{3}z$ $z \geqslant 35$ $\because x \geqslant 0$ $\frac{200}{3}-\frac{5}{3}z \geqslant 0$ $z \leqslant 40$ $\therefore 35 \leqslant z \leqslant 40$ x, y, z are non-negative integers. Using GC table: $z=37 \Rightarrow x=5,\ y=58$ $z=40 \Rightarrow x=0,\ y=60$ $\therefore (x,y,z)=(5,58,37)$ or $(0,60,40)$	

2 Inequalities

2.1 **(a)** Solve the inequality $|e^x - 2| > \dfrac{1}{x^2 - 1}$ using a graphical approach. [3]

(b) Without the use of a graphing calculator, solve the inequality $\dfrac{x+2}{x-1} \geqslant \dfrac{2}{x-3}$. [3]

[RVHS 2026 Prelim P1 Q2]

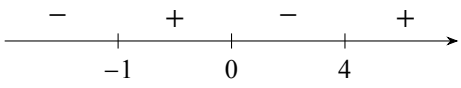
Solution		Comments
(a) **[3]**	 Using GC, intersections at $x=-1.2581$, $x=1.2770$ $\therefore x<-1.26$ or $-1<x<1$ or $x>1.28$	
(b) **[3]**	$\dfrac{x+2}{x-1} - \dfrac{2}{x-3} \geqslant 0$ $\dfrac{(x+2)(x-3)-2(x-1)}{(x-1)(x-3)} \geqslant 0$ $\dfrac{x^2-3x-4}{(x-1)(x-3)} \geqslant 0$ $\dfrac{(x-4)(x+1)}{(x-1)(x-3)} \geqslant 0$  $\therefore x \leqslant -1$ or $1<x<3$ or $x \geqslant 4$	

2.2 Use an algebraic method to solve the inequalities

$$\frac{x(5x+9)}{x^2+2x+5} > x \geqslant 3x-6.$$

[4]

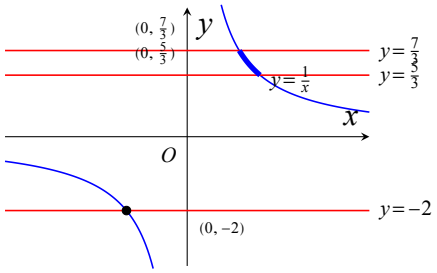
[ACJC 2026 Prelim P1 Q1]

Solution		Comments
[4]	$x \geqslant 3x-6$ $x \leqslant 3 \quad ---(1)$ $x^2+2x+5=(x+1)^2+4>0$ for all real x $\frac{x(5x+9)}{x^2+2x+5} > x$ $5x^2+9x > x(x^2+2x+5) \quad (\because x^2+2x+5>0)$ $x^3-3x^2-4x<0$ $x(x-4)(x+1)<0$  $x<-1$ or $0<x<4 \quad ---(2)$ $(1) \cap (2)$: $\therefore x<-1$ or $0<x \leqslant 3$	

2.3 **(i)** Solve the inequality $3\,|x^2-4| \leqslant |x+2|$. [4]

(ii) Hence, solve $\dfrac{3}{x^2}\,|1-4x^2| \leqslant \left|\dfrac{1+2x}{x}\right|$. [2]

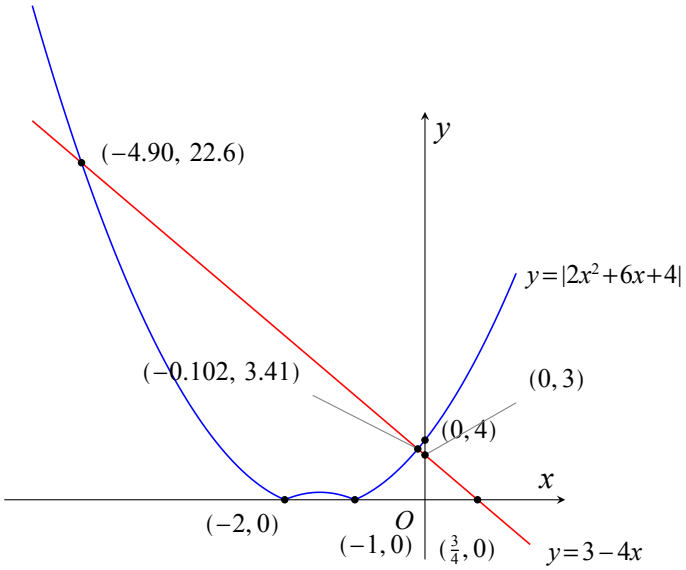
[SAJC 2025 Prelim P2 Q1]

Solution		Comments
(i) **[4]**	$3\|x-2\|\|x+2\| \leqslant \|x+2\|$ $x=-2:\quad 0 \leqslant 0$ (true) $x \neq -2:\ \|x+2\|>0$ $3\|x-2\| \leqslant 1$ $-\frac{1}{3} \leqslant x-2 \leqslant \frac{1}{3}$ $\frac{5}{3} \leqslant x \leqslant \frac{7}{3}$ $\therefore x=-2$ or $\frac{5}{3} \leqslant x \leqslant \frac{7}{3}$	
(ii) **[2]**	$\dfrac{3}{x^2}\,\|1-4x^2\| = 3\left\|\dfrac{1}{x^2}-4\right\| \quad (\because x^2>0)$ $\left\|\dfrac{1+2x}{x}\right\| = \left\|\dfrac{1}{x}+2\right\|$ $3\left\|\left(\dfrac{1}{x}\right)^2-4\right\| \leqslant \left\|\dfrac{1}{x}+2\right\| \quad (x \to \frac{1}{x})$ $\dfrac{1}{x}=-2$ or $\dfrac{5}{3} \leqslant \dfrac{1}{x} \leqslant \dfrac{7}{3}$  $\therefore x=-\frac{1}{2}$ or $\frac{3}{7} \leqslant x \leqslant \frac{3}{5}$	

2.4 **(i)** On the same axes, sketch the curves with equations $y=|2x^2+6x+4|$ and $y=3-4x$, indicating any intercepts with the axes and points of intersection. Hence solve the inequality $3-4x<|2x^2+6x+4|$. [4]

(ii) Find the exact area bounded by the graphs of $y=3-4x$, $y=|2x^2+6x+4|$, $x=-3$ and $x=-1$. [3]

[SAJC 2019 Prelim P1 Q2]

Solution		**Comments**
(i) **[4]**		
	$3-4x=2x^2+6x+4$ $2x^2+10x+1=0$ $x=\dfrac{-5\pm\sqrt{23}}{2}$ $\therefore x<\dfrac{-5-\sqrt{23}}{2}$ or $x>\dfrac{-5+\sqrt{23}}{2}$	
(ii) **[3]**	$\text{Area}=\int_{-3}^{-1}(3-4x-\lvert2x^2+6x+4\rvert)\,dx$ $=\int_{-3}^{-2}(3-4x-2x^2-6x-4)\,dx+\int_{-2}^{-1}(3-4x+2x^2+6x+4)\,dx$ $=\int_{-3}^{-2}(-2x^2-10x-1)\,dx+\int_{-2}^{-1}(2x^2+2x+7)\,dx$ $=\left[-\frac{2}{3}x^3-5x^2-x\right]_{-3}^{-2}+\left[\frac{2}{3}x^3+x^2+7x\right]_{-2}^{-1}$ $=\left(\frac{16}{3}-20+2\right)-(18-45+3)+\left(-\frac{2}{3}+1-7\right)-\left(-\frac{16}{3}+4-14\right)$ $=20$	

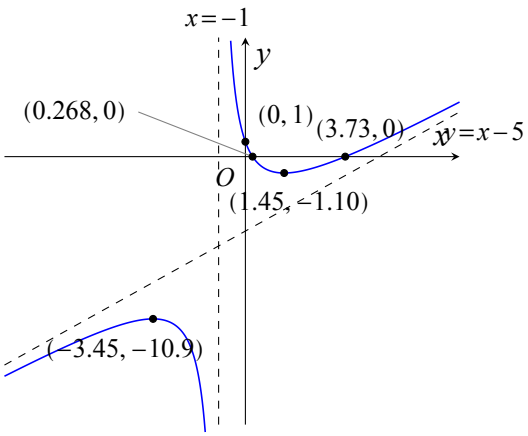
3 Graphing Techniques

3.1 A curve C has equation $y=\dfrac{x^2-4x+1}{x+1}$, $x\neq -1$.

(a) Prove, using an algebraic method, that $y=\dfrac{x^2-4x+1}{x+1}$ cannot lie between two certain values, to be determined in exact form. [3]

(b) Sketch C, stating the equations of any asymptotes, and the coordinates of any turning points and any axial intercepts. [4]

[VJC 2025 Promo Q9(a), (b)]

Solution		Comments
(a) **[3]**	$y=\dfrac{x^2-4x+1}{x+1}$ $yx+y=x^2-4x+1$ $x^2+(-y-4)x+(1-y)=0$ No real solution for x: $(-y-4)^2-4(1-y)<0$ $y^2+8y+16-4+4y<0$ $y^2+12y+12<0$ $(y+6)^2-24<0$ $\left(y+6-\sqrt{24}\right)\left(y+6+\sqrt{24}\right)<0$ $-6-2\sqrt{6}<y<-6+2\sqrt{6}$ $\therefore y$ cannot lie between $-6-2\sqrt{6}$ and $-6+2\sqrt{6}$	
(b) **[4]**	$y=x-5+\dfrac{6}{x+1}$ Asymptotes: $x=-1,\ y=x-5$ $x=0$: $y=1$ $y=0$: $x^2-4x+1=0 \Rightarrow x=0.268,\ 3.73$ Turning points: $(1.45,-1.10),\ (-3.45,-10.9)$  	

3.2 A curve C has equation $y=ax+b+\dfrac{2b-a}{x+3}$, where a and b are constants such that $a>0$, $b\neq\dfrac{1}{2}a$ and $x\neq -3$.

(a) If C has no stationary points, use differentiation to find the relationship between a and b. [3]

(b) It is now given that $b=a$.

(i) Find the x-coordinates of the turning points. [2]

(ii) Hence sketch C, stating the equations of any asymptotes and the coordinates of the axial intercepts and turning points. [3]

(iii) Verify that the point of intersection of the two asymptotes of C lies on the line $y=kx+3k-2a$.

Hence, using the graph in part (b)(ii), find the range of values of k in terms of a, such that the equation $x(k-a)+3k-2a-b-\dfrac{2b-a}{x+3}=0$ has real roots. [2]

[SAJC 2025 Promo Q7]

Solution		Comments
(a) **[3]**	$\dfrac{dy}{dx}=a-\dfrac{2b-a}{(x+3)^2}$ $\dfrac{dy}{dx}=0:\quad a-\dfrac{2b-a}{(x+3)^2}=0$ $(x+3)^2=\dfrac{2b-a}{a}\qquad ---(1)$ No stationary points $\Rightarrow \dfrac{2b-a}{a}<0$ Since $a>0,\quad 2b-a<0$ $\therefore a>2b$	
(b)(i) **[2]**	$b=a:\quad (x+3)^2=\dfrac{2a-a}{a}=1$ $x+3=1$ or -1 $x=-2$ or -4	

(b)(ii)
[3]

$$y = ax + a + \frac{a}{x+3}$$

Asymptotes: $x = -3,\ y = ax + a$

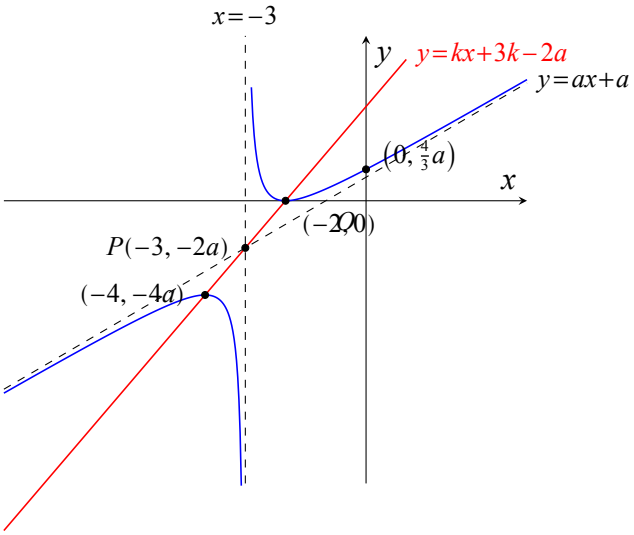
$x = -2 : y = 0; \quad x = -4 : y = -4a$

Turning points: $(-2, 0),\ (-4, -4a)$

$$x = 0 : y = \frac{4}{3}a$$

$$y = 0 : ax + a + \frac{a}{x+3} = 0 \Rightarrow (x+1)(x+3) + 1 = 0$$

$$x^2 + 4x + 4 = 0 \Rightarrow (x+2)^2 = 0 \Rightarrow x = -2$$



(b)(iii)
[2]

$x = -3$ and $y = ax + a$: $\quad y = -3a + a = -2a$

Intersection of asymptotes: $P(-3, -2a)$

$x = -3$: $\quad \text{RHS} = k(-3) + 3k - 2a = -2a = \text{LHS}$

$\therefore P$ lies on the line $y = kx + 3k - 2a$

$$x(k-a) + 3k - 2a - b - \frac{2b-a}{x+3} = 0$$

$$kx + 3k - 2a = ax + b + \frac{2b-a}{x+3}$$

Line through P meets $C \Rightarrow k > a$

3.3 A curve C has equation $\mathrm{f}(x)=\dfrac{3x^2+6x+k}{x+2}$, where $x>-2$ and k is a non-zero real constant.

It is given that the function f is strictly increasing.

(a) Find the range of values of k. [3]

(b) Sketch the curve C, giving the coordinates of any points where C crosses the x- and y-axes and the equations of any asymptotes. [3]

(c) By adding a suitable curve to the graph of the curve C in (b), deduce the number of distinct real root(s) of the equation $(3x^2+6x+k)^2=(x^2+1)\,(9x^2+36x+36)$. [3]

[DHS 2025 Promo Q8]

Solution		Comments
(a) **[3]**	$\mathrm{f}(x)=\dfrac{3x^2+6x+k}{x+2}=3x+\dfrac{k}{x+2}$	
	$\mathrm{f}'(x)=3-\dfrac{k}{(x+2)^2}=\dfrac{3x^2+12x+12-k}{(x+2)^2}$	
	f strictly increasing $\Rightarrow$ $\mathrm{f}'(x)>0$	
	$3x^2+12x+12-k>0$ $\quad\because (x+2)^2>0$ for $x>-2$	
	$3(x+2)^2-k>0$	
	Since $3(x+2)^2-k>-k$ for all $x>-2$, $\quad -k\geqslant 0$	
	$k\neq 0 \Rightarrow k<0$	
(b) **[3]**	$y=3x+\dfrac{k}{x+2}$	
	Asymptotes: $y=3x,\ x=-2$	
	$x=0:\ y=\dfrac{k}{2}$	
	$y=0:\ 3x^2+6x+k=0 \Rightarrow x=\dfrac{-6\pm\sqrt{36-12k}}{6}=-1\pm\dfrac{\sqrt{9-3k}}{3}$	
	$x>-2 \Rightarrow x=-1+\dfrac{\sqrt{9-3k}}{3}$	

(c)
[3]

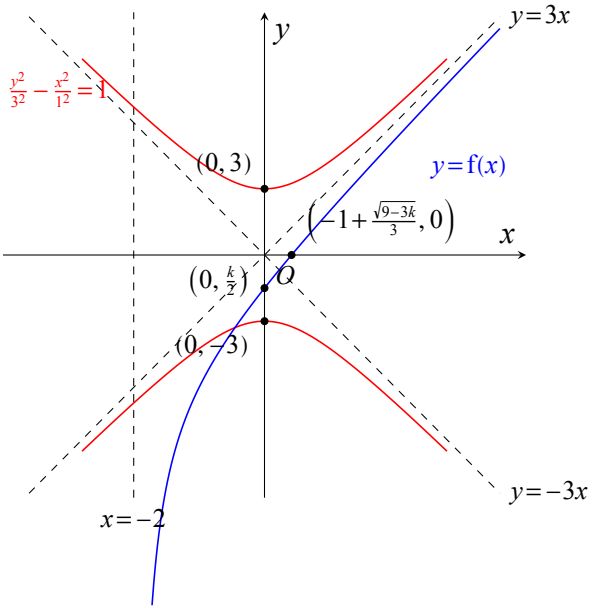
$$(3x^2+6x+k)^2=(x^2+1)\,(9x^2+36x+36) \quad ---(*)$$

$$(3x^2+6x+k)^2=9\,(x^2+1)\,(x+2)^2$$

$$\left(\frac{3x^2+6x+k}{x+2}\right)^2-9x^2=9$$

Draw the hyperbola $\dfrac{y^2}{3^2}-\dfrac{x^2}{1^2}=1$

Asymptotes $y=\pm 3x$, vertices $(0,\pm 3)$



$y=\mathrm{f}(x)$ and $\dfrac{y^2}{3^2}-\dfrac{x^2}{1^2}=1$ have one intersection point

$\therefore$ $(*)$ has one real root

3.4 The curve C is defined by the parametric equations $x=6t-5$ and $y=2t^2+1$, where $t \geqslant \frac{5}{6}$.

(a) Sketch the curve C, labelling the exact coordinates of any axial intercept(s). [1]

(b) Find the equation of the normal to C at the point P where $t=1$. [2]

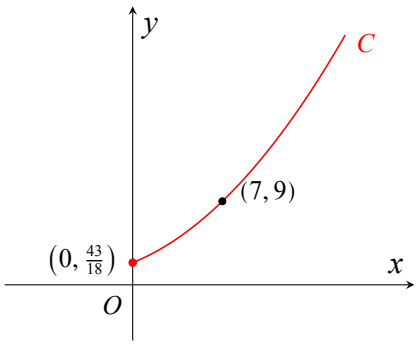
(c) Find the acute angle between the normal to C at the point P where $t=1$ and the tangent to C at the point M where $t=3$. [2]

(d) The curve C is translated 5 units in the positive x-direction and translated 3 units in the negative y-direction, to form the curve D. Find the equation of D in parametric form. [2]

The curve E is defined by the parametric equations $x=7u$ and $y=\frac{9}{u}$, where $u \neq 0$.

(e) Show that at the point of intersection of the curves C and E, $6t^3-5t^2+3t-34=0$. Deduce that there is only one point of intersection and find the coordinates of this point. [4]

[RVHS 2025 Prelim P1 Q12]

Solution		**Comments**
(a) **[1]**	$t=\frac{5}{6}$: $x=6\left(\frac{5}{6}\right)-5=0$, $y=2\left(\frac{5}{6}\right)^2+1=\frac{43}{18}$ Endpoint (y-intercept): $\left(0, \frac{43}{18}\right)$  	

(b) **[2]**

$$\frac{dx}{dt}=6, \quad \frac{dy}{dt}=4t$$

$$\frac{dy}{dx}=\frac{4t}{6}=\frac{2}{3}t$$

$t=1$: $\quad \frac{dy}{dx}=\frac{2}{3} \Rightarrow$ gradient of normal $=-\frac{3}{2}$

$t=1$: $\quad x=1,\ y=3$

$$y-3=-\frac{3}{2}(x-1)$$

$$y=-\frac{3}{2}x+\frac{9}{2}$$

(c) **[2]**

$t=3$: $\quad$ gradient of tangent $=\frac{2}{3}(3)=2$

$$\theta_1+\theta_2=\tan^{-1}(2)-\tan^{-1}\left(-\frac{3}{2}\right)=2.089942 \text{ or } 119.7^\circ$$

$$\text{Acute angle}=\pi-2.089942 \text{ or } 180^\circ-119.7^\circ$$

$$=1.05 \text{ or } 60.3^\circ$$

(d) **[2]**

$x-5=6t-5 \Rightarrow x=6t$

$y+3=2t^2+1 \Rightarrow y=2t^2-2$

D: $\quad x=6t,\ y=2t^2-2$

(e) **[4]**

$$6t-5=7u \Rightarrow u=\frac{6t-5}{7} \quad ---(1)$$

$$2t^2+1=\frac{9}{u} \quad ---(2)$$

Sub (1) into (2): $2t^2+1=\frac{63}{6t-5}$

$$(2t^2+1)(6t-5)=63$$

$$12t^3-10t^2+6t-68=0$$

$$6t^3-5t^2+3t-34=0 \quad \text{(shown)}$$

$$(t-2)(6t^2+7t+17)=0$$

$6t^2+7t+17$: $b^2-4ac=7^2-4(6)(17)=-359<0 \Rightarrow$ no real roots

$\therefore t=2$ only $\Rightarrow$ only one point of intersection

$t=2$: $\quad x=6(2)-5=7,\ y=2(2)^2+1=9$

Point of intersection: $(7, 9)$

3.5 The parametric equations of the curve C are

$$x=\tan\theta-1 \quad \text{and} \quad y=2\sec\theta, \quad \text{where } -\frac{\pi}{2}<\theta<\frac{\pi}{2}.$$

(a) Find $\frac{\mathrm{d}y}{\mathrm{d}x}$ in terms of θ, leaving your answer in a single trigonometric function. [2]

(b) Find the equation of the tangent to C at the point where it cuts the y-axis, leaving your answer in exact form. [3]

(c) Find the Cartesian equation of C. [1]

(d) Find the equations of the asymptotes and state the coordinates of any turning point(s). [3]

(e) The region bounded by the tangent found in part (b), the curve C and the line $x=-1$ is rotated completely about the x-axis. Calculate the volume obtained. [3]

[HCI 2024 Promo Q12]

Solution		Comments
(a) **[2]**	$\frac{\mathrm{d}y}{\mathrm{d}x}=\frac{\mathrm{d}y}{\mathrm{d}\theta}\div\frac{\mathrm{d}x}{\mathrm{d}\theta}=\frac{2\sec\theta\tan\theta}{\sec^2\theta}=2\sin\theta$	
(b) **[3]**	$x=0:\quad \tan\theta-1=0 \Rightarrow \tan\theta=1$ $\theta=\frac{\pi}{4} \quad \text{since } -\frac{\pi}{2}<\theta<\frac{\pi}{2}$ $\theta=\frac{\pi}{4}:\quad y=\frac{2}{\cos\frac{\pi}{4}}=2\sqrt{2}, \quad \frac{\mathrm{d}y}{\mathrm{d}x}=2\sin\frac{\pi}{4}=\sqrt{2}$ Tangent: $y=\sqrt{2}x+2\sqrt{2}$	
(c) **[1]**	$\tan^2\theta+1=\sec^2\theta$ $(x+1)^2+1=\left(\frac{y}{2}\right)^2$ $-\frac{\pi}{2}<\theta<\frac{\pi}{2} \Rightarrow 0<\cos\theta\leqslant 1 \Rightarrow \sec\theta\geqslant 1$ $\therefore (x+1)^2+1=\left(\frac{y}{2}\right)^2, \quad y\geqslant 2$	

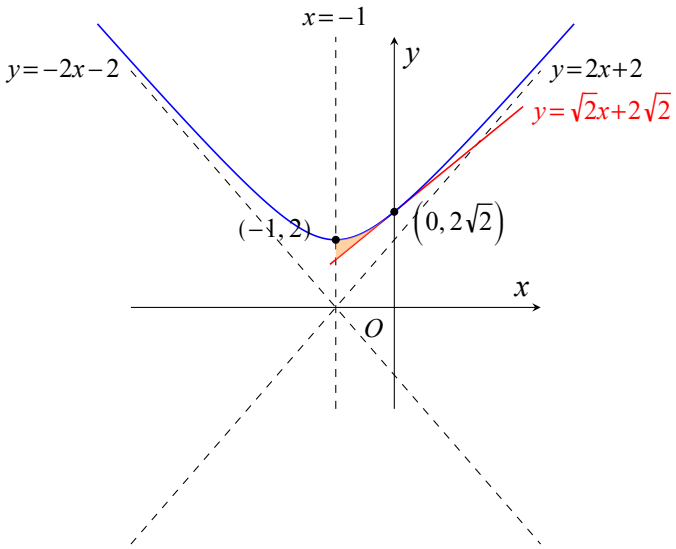
(d)
[3]

Asymptotes: $\left(\frac{y}{2}\right)^2 = (x+1)^2$

$$\frac{y}{2} = \pm(x+1)$$

$y = 2x+2$ or $y = -2x-2$

Centre $(-1, 0)$, $y \geqslant 2 \Rightarrow$ turning point $(-1, 2)$



(e)
[3]

$$\frac{y^2}{4} = (x+1)^2 + 1$$

$$y^2 = 4(x+1)^2 + 4$$

$$\text{Volume} = \pi \int_{-1}^{0} \left[4\left((x+1)^2+1\right) - \left(\sqrt{2}x + 2\sqrt{2}\right)^2 \right] dx$$

$$= 2.09 \text{ units}^3$$

4 Graphing Transformations

4.1 A curve with equation $y=\mathrm{h}(x)$ undergoes the following sequence of transformations.

A: A translation of 2 units in the positive direction of the x-axis.

B: A scaling parallel to the x-axis by factor of 3.

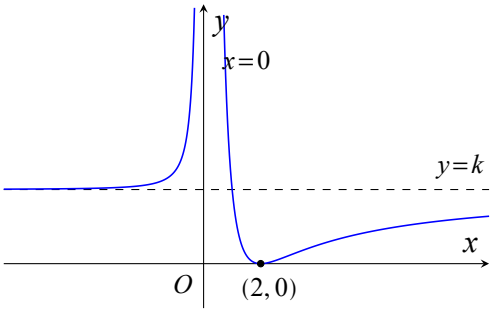
C: A reflection in the y-axis.

The resultant curve has equation $y=-\dfrac{x^2}{6x+45}$. Find $\mathrm{h}(x)$. [3]

[NJC 2024 Promo Q4(b)]

Solution		Comments
[3] **(b)**	$y=-\dfrac{x^2}{6x+45}$ C^{-1}: reflection in the y-axis $(x\to -x)$ $y=-\dfrac{(-x)^2}{6(-x)+45}=-\dfrac{x^2}{-6x+45}$ B^{-1}: scaling parallel to the x-axis by factor $\frac{1}{3}$ $(x\to 3x)$ $y=-\dfrac{(3x)^2}{-6(3x)+45}=-\dfrac{9x^2}{-18x+45}=-\dfrac{x^2}{-2x+5}$ A^{-1}: translation of 2 units in the negative x-direction $(x\to x+2)$ $y=-\dfrac{(x+2)^2}{-2(x+2)+5}=-\dfrac{(x+2)^2}{-2x+1}$ $\mathrm{h}(x)=\dfrac{(x+2)^2}{2x-1}$	

4.2 The diagram below shows the graph of $y=\mathrm{f}(2x)$. The graph has a turning point at $(2,0)$, and asymptotes with equations $x=0$ and $y=k$.



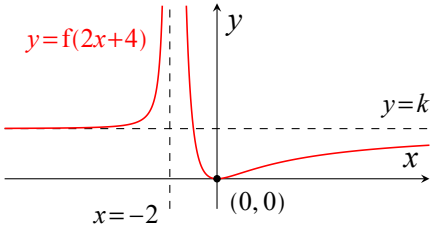
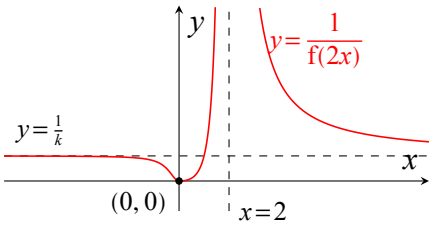
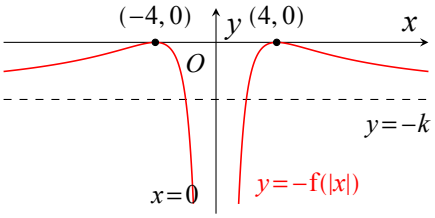
(a) State a single transformation that will transform the graph of $y=\mathrm{f}(2x)$ onto the graph of $y=\mathrm{f}(2x+4)$. Hence sketch the graph of $y=\mathrm{f}(2x+4)$. [3]

On separate clearly labelled diagrams, sketch the graphs of

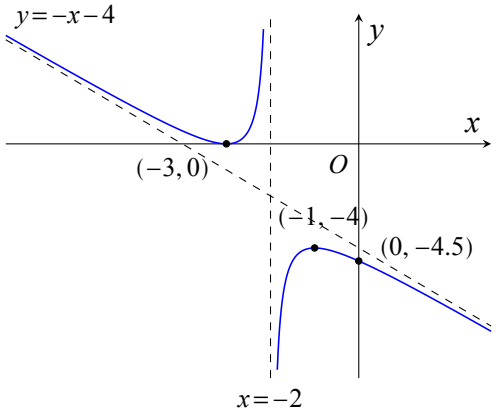
(b) $y=\dfrac{1}{\mathrm{f}(2x)}$, [3]

(c) $y=-\mathrm{f}(|x|)$. [3]

[EJC 2025 Prelim P1 Q4]

Solution		Comments
(a) **[3]**		
(b) **[3]**		
(c) **[3]**		

4.3



The diagram shows the curve $y = \mathrm{f}(x)$. The curve crosses the x-axis at its minimum point $(-3, 0)$. It has a maximum turning point $(-1, -4)$ and crosses the y-axis at the point $(0, -4.5)$. The asymptotes of the curve are $x = -2$ and $y = -x - 4$.

On separate diagrams, sketch the graphs of

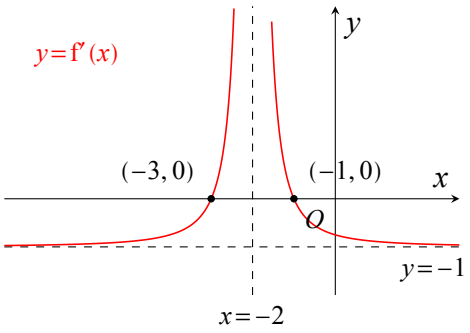
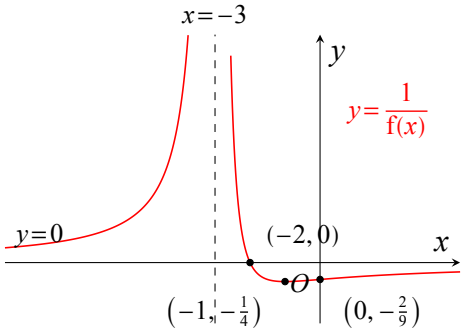
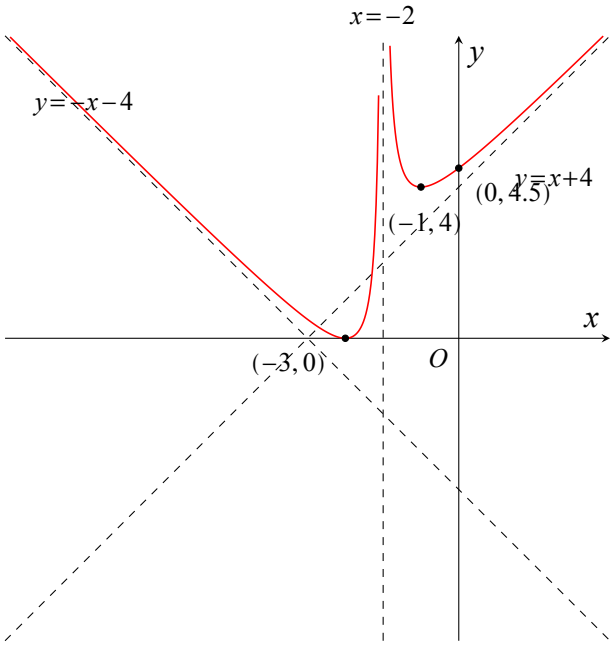
(a) $y = \mathrm{f}'(x)$, [3]

(b) $y = \dfrac{1}{\mathrm{f}(x)}$, [3]

(c) $y = |\mathrm{f}(x)|$, [3]

indicating the equations of any asymptotes, turning points and intersections with the axes.

[CJC 2025 Promo Q8]

Solution	Comments
(a) **[3]** 	
(b) **[3]** 	
(c) **[3]** 	

4.4 Express $\frac{9x+5}{3x+2}$ in the form $a-\frac{b}{3x+2}$ where a and b are positive constants.

Hence, describe a sequence of transformations which transform the graph of $y=\frac{9x+5}{3x+2}$ to the graph of $y=\frac{1}{x}$. [5]

[RVHS 2026 Prelim P1 Q10(b)]

Solution		**Comments**
[5] **(b)**	$\frac{9x+5}{3x+2}=3-\frac{1}{3x+2} \quad (a=3,\ b=1)$ $y=3-\frac{1}{3x+2} \xrightarrow{y\to y+3} y=-\frac{1}{3x+2}$ $\xrightarrow{y\to -y} y=\frac{1}{3x+2}$ $\xrightarrow{x\to \frac{x}{3}} y=\frac{1}{x+2}$ $\xrightarrow{x\to x-2} y=\frac{1}{x}$ 1. Translate 3 units in the negative y-direction 2. Reflect about the x-axis 3. Scale by factor 3 parallel to the x-axis 4. Translate 2 units in the positive x-direction	

5 Functions

5.1 The function f is defined by

$$\mathrm{f}: x \to \left|\frac{2x+4}{3-x}\right|, \qquad x \in \mathbb{R}, \qquad x \neq 3.$$

(a) Sketch the graph of $y = \mathrm{f}(x)$, giving the equations of any asymptotes and the coordinates of the points where the curve meets the axes. [3]

(b) Hence state the range of f. [1]

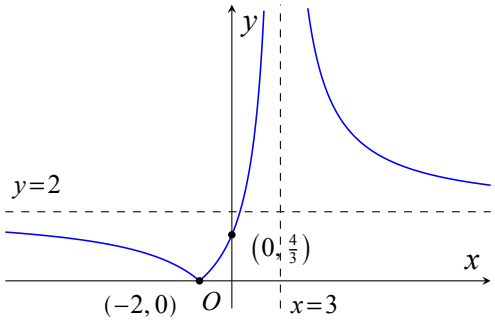
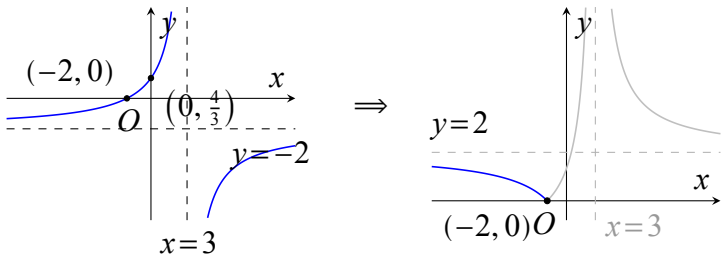
(c) Explain why the function f^2 does not exist. [1]

The domain of f is further restricted to $x \leqslant a$, where a is a constant.

(d) State the greatest value of a such that the function f^{-1} exists. [1]

(e) Hence find $\mathrm{f}^{-1}(x)$ and state its domain. [4]

[2023 A-Level P1 Q7]

Solution		Comments
(a) **[3]**	$y=\left\|\frac{2x+4}{3-x}\right\|=\left\|-2+\frac{10}{3-x}\right\|$ Asymptotes: $y=\|-2\|=2, \quad x=3$ $x=0: \quad y=\left\|\frac{4}{3}\right\|=\frac{4}{3}$ $y=0: \quad x=-2$  	
(b) **[1]**	$\mathrm{R_f}=[0,\infty)$	
(c) **[1]**	$\mathrm{R_f}=[0,\infty), \quad \mathrm{D_f}=(-\infty,3)\cup(3,\infty)$ $3\in\mathrm{R_f},\ 3\notin\mathrm{D_f} \Rightarrow \mathrm{R_f}\not\subseteq\mathrm{D_f}$ $\therefore$ f^2 does not exist	
(d) **[1]**	$a=-2$	
(e) **[4]**	$x\leqslant -2: \quad \mathrm{f}(x)=\left\|\frac{2x+4}{3-x}\right\|=-\frac{2x+4}{3-x}$  $y=\mathrm{f}(x)$ $-3y+xy=2x+4$ $x(y-2)=4+3y$ $x=\frac{4+3y}{y-2}$ $\mathrm{f}^{-1}(x)=\frac{4+3x}{x-2}, \qquad \mathrm{D_{f^{-1}}}=[0,2)$	

5.2 The function f is defined by

$$\mathrm{f}: x \mapsto \frac{ax-3}{5x-4}, \text{ for } x \in \mathbb{R}, \ x \neq \frac{4}{5}, \text{ and } a \in \mathbb{R}.$$

(a) Find the range of f in terms of a. [1]

The function f is such that $\mathrm{f}(x) = \mathrm{f}^{-1}(x)$ for all x in the domain of f.

(b) Find the value of a. [2]

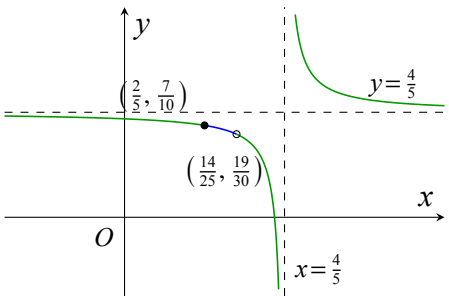
(c) Hence find $\mathrm{f}^{2025}(2)$. [2]

Another function g is defined by

$$\mathrm{g}: x \mapsto \left(x - \frac{1}{5}\right)^2 + \frac{2}{5}, \text{ for } x \in \mathbb{R}, \ 0 < x < \frac{3}{5}.$$

(d) Find the range of fg. [2]

[HCI 2025 Prelim P1 Q8]

Solution		Comments
(a) **[1]**	Horizontal asymptote: $y=\frac{a}{5}$ $\mathrm{R_f}=\mathbb{R}\setminus\left\{\frac{a}{5}\right\}$	
(b) **[2]**	$y=\frac{ax-3}{5x-4}$ $5xy-4y=ax-3$ $5xy-ax=4y-3$ $x=\frac{4y-3}{5y-a}$ $\mathrm{f}^{-1}(x)=\frac{4x-3}{5x-a},\quad x\in\mathbb{R},\ x\neq\frac{a}{5}$ $\mathrm{f}(x)=\mathrm{f}^{-1}(x)\ \Rightarrow\ a=4$	
(c) **[2]**	$\mathrm{f}^{-1}(x)=\mathrm{f}(x)\ \Rightarrow\ \mathrm{f}^2(x)=\mathrm{ff}^{-1}(x)=x$ $\mathrm{f}^{2025}(2)=\mathrm{f}^2\left(\mathrm{f}^{2023}(2)\right)$ $=\mathrm{f}^2\left(\mathrm{f}^{2021}(2)\right)$ $=\cdots$ $=\mathrm{f}(2)$ $=\frac{4(2)-3}{5(2)-4}=\frac{5}{6}$	
(d) **[2]**	$\mathrm{R_g}=\left[\frac{2}{5},\frac{14}{25}\right)$ $\mathrm{f}(x)=\frac{4x-3}{5x-4}$ $\left(0,\frac{3}{5}\right)\xrightarrow{\mathrm{g}}\left[\frac{2}{5},\frac{14}{25}\right)\xrightarrow{\mathrm{f}}\left(\frac{19}{30},\frac{7}{10}\right]$ $\mathrm{R_{fg}}=\left(\frac{19}{30},\frac{7}{10}\right]$  	

5.3 The function f is given by

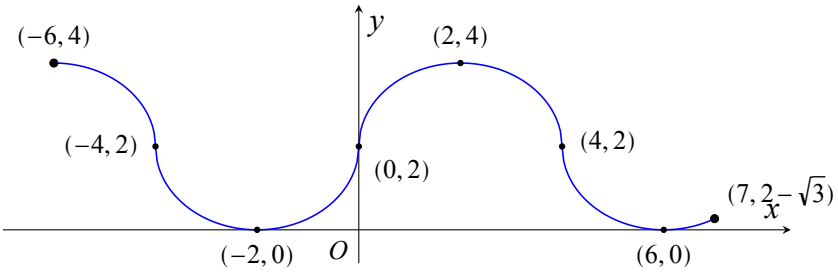
$$\mathrm{f}(x)=\begin{cases}2+\sqrt{2^2-(x-2)^2}, & 0<x\leqslant 4,\\ 2-\sqrt{2^2-(x-6)^2}, & 4<x\leqslant 8.\end{cases}$$

It is given that $\mathrm{f}(x)=\mathrm{f}(x+8)$ for all real values of x.

(a) On the diagram in the Printed Answer Book, sketch the graph of $y=\mathrm{f}(x)$ for $-6\leqslant x\leqslant 7$, indicating clearly the coordinates of the end points and the points where the graph cuts the axes. [3]

(b) Without integrating, write down the exact area of the region bounded by $y=\mathrm{f}(x)$, the line $x=4$, the x-axis and the y-axis. [1]

[VJC 2025 Prelim P1 Q5(a), (b)]

Solution		Comments
(a) **[3]**	$y=2+\sqrt{2^2-(x-2)^2}$ $y-2=\sqrt{2^2-(x-2)^2}$ $(x-2)^2+(y-2)^2=2^2$ Centre $(2,2)$, radius 2. Since it is a positive square root, it is the top half of this circle. $y=2-\sqrt{2^2-(x-6)^2}$ $y-2=-\sqrt{2^2-(x-6)^2}$ $(x-6)^2+(y-2)^2=2^2$ Centre $(6,2)$, radius 2. Since it is a negative square root, it is the bottom half of this circle.  	
(b) **[1]**	$\text{Area}=2\times 4+\frac{1}{2}\pi(2^2)=8+2\pi$	

5.4 The function f is defined by

$$\mathrm{f}(x) = 1 - ax,\ x \in \mathbb{R},\ \text{where } a \text{ is a real constant.}$$

The function g is defined by

$$\mathrm{g}(x) = \begin{cases} x^2 + 3 & \text{for} \quad 0 \leqslant x \leqslant 2, \\ 7 - x & \text{for} \quad 2 < x \leqslant 7. \end{cases}$$

(i) Find the set of possible values of a such that f^{-1} exist. [1]

(ii) If $a = 2$, describe a sequence of transformations that transform the graph of $y = \mathrm{g}(x)$ onto the graph of $y = \mathrm{fg}(x)$. [3]

(iii) The function gg is defined by

$$\mathrm{gg}(x) = \begin{cases} \mathrm{h}_1(x) & \text{for} \quad 0 \leqslant x \leqslant 2, \\ \mathrm{h}_2(x) & \text{for} \quad 2 < x < 5, \\ \mathrm{h}_3(x) & \text{for} \quad 5 \leqslant x \leqslant 7. \end{cases}$$

Find $\mathrm{h}_1(x)$, $\mathrm{h}_2(x)$ and $\mathrm{h}_3(x)$. [3]

[ACJC 2022 Prelim P2 Q1]

Solution		Comments
(i) **[1]**	$a \in \mathbb{R} \setminus \{0\}$	
(ii) **[3]**	$y = \mathrm{fg}(x) = 1 - 2\mathrm{g}(x)$ Reflection of the graph of $y = \mathrm{g}(x)$ in the x-axis, followed by scaling by scale factor 2 parallel to the y-axis, followed by translation 1 unit in the positive y-direction. **OR** Translation of the graph of $y = \mathrm{g}(x)$ by $\frac{1}{2}$ unit in the negative y-direction followed by reflection in the x-axis followed by scaling by scale factor 2 parallel to the y-axis. **OR** Translation of the graph of $y = \mathrm{g}(x)$ by $\frac{1}{2}$ unit in the negative y-direction followed by scaling by scale factor 2 parallel to the y-axis followed by reflection in the x-axis.	
(iii) **[3]**	$\mathrm{gg}(x) = \begin{cases} 7-(x^2+3) & \text{for} \quad 0 \leqslant x \leqslant 2 \\ 7-(7-x) & \text{for} \quad 2 < x < 5 \\ (7-x)^2+3 & \text{for} \quad 5 \leqslant x \leqslant 7 \end{cases}$ $\therefore \mathrm{gg}(x) = \begin{cases} 4-x^2 & \text{for} \quad 0 \leqslant x \leqslant 2 \\ x & \text{for} \quad 2 < x < 5 \\ x^2-14x+52 & \text{for} \quad 5 \leqslant x \leqslant 7 \end{cases}$	

5.5 Functions f and g are defined by

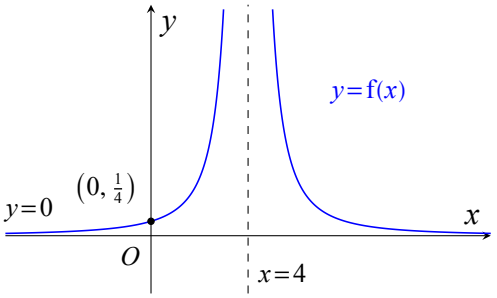
$$\mathrm{f}: x \to \frac{4}{(x-4)^2}, \quad x \in \mathbb{R},\ x \neq 4,$$

$$\mathrm{g}: x \to \ln\left(1+\frac{1}{x}\right), \quad x \in \mathbb{R},\ x > 0.$$

(a) Sketch the graph of $y = \mathrm{f}(x)$, stating the equations of any asymptotes, the coordinates of the points where it crosses the axes and the coordinates of the turning points, if any. [2]

(b) Show that gf exists. Hence find the rule, domain and range of gf. [4]

[CJC 2025 Prelim P2 Q2(a), (b)]

Solution		Comments
(a) **[2]**		

(b)
[4]

For gf to exist, $R_f \subseteq D_g$,

$R_f = (0, \infty)$

$D_g = (0, \infty)$

Since $R_f \subseteq D_g$, gf exists (shown)

$$\begin{aligned} gf(x) &= g\left[\frac{4}{(x-4)^2}\right] \\ &= \ln\left[1+\left(\frac{1}{\frac{4}{(x-4)^2}}\right)\right] \\ &= \ln\left[1+\frac{(x-4)^2}{4}\right] \end{aligned}$$

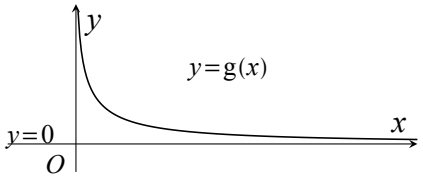
$D_{gf} = D_f = (-\infty, 4) \cup (4, \infty)$

OR

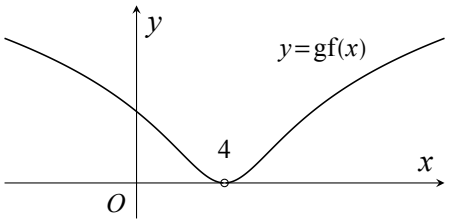
$D_{gf} = D_f = \mathbb{R} \setminus \{4\}$

Method 1: Mapping

$D_f = (-\infty, 4) \cup (4, \infty) \xrightarrow{f} R_f = (0, \infty) \xrightarrow{g} R_{gf} = (0, \infty)$



Method 2: Sketch $y = gf(x)$ for given domain



$R_{gf} = (0, \infty)$

5.6 The function g is defined by $g(0)=g(1)=1$, $g(2n)=2+g(n)$ and $g(2n+1)=1+g(2n)$, for integer n.

(i) Find $g(4)$, $g(7)$ and $g(12)$.

(ii) Does g have an inverse? Justify your answer. [3]

[2016 A-Level P1 Q10(b)]

Solution		**Comments**
(i)	$g(4)=2+g(2)$ $=2+2+g(1)$ $=2+2+1+g(0)$ $=2+2+1+1$ $=6$ $g(7)=1+g(6)$ $=1+2+g(3)$ $=1+2+1+g(2)$ $=1+2+1+2+1+1$ $=8$ $g(12)=2+g(6)$ $=2+2+1+2+1+1$ $=9$	
(ii) **[3]**	$g(8)=2+g(4)$ $=2+6$ $=8$ $=g(7)$ Since $g(8)=g(7)=8$, g is not one-one, so g^{-1} does not exist.	

6 Sequences and Series

6.1 It is given $\sum_{r=1}^{n} r^3 = \frac{n^2(n+1)^2}{4}$ and $u_r = 2r^3 + 5$.

(i) Find $\sum_{r=1}^{n} u_r$. [2]

(ii) Hence, or otherwise, find $\sum_{r=2}^{n} (2(r+2)^3 + 5)$. [3]

(iii) Explain why the series $\sum_{r=1}^{\infty} u_r$ does not converge. [1]

[JPJC 2025 Prelim P1 Q8(b)]

Solution		Comments
(i) **[2]**	$\sum_{r=1}^{n}(2r^3+5) = 2\sum_{r=1}^{n} r^3 + \sum_{r=1}^{n} 5$ $= 2\left(\frac{n^2(n+1)^2}{4}\right) + 5n$ $= \frac{n^2(n+1)^2}{2} + 5n$	
(ii) **[3]**	$\sum_{r=2}^{n}(2(r+2)^3+5) = (2(2+2)^3+5) + \cdots + (2(n+2)^3+5)$ $= \sum_{r=4}^{n+2}(2r^3+5)$ $= \sum_{r=1}^{n+2}(2r^3+5) - \sum_{r=1}^{3}(2r^3+5)$ $= \frac{(n+2)^2(n+3)^2}{2} + 5(n+2) - \left[\left(\frac{3^2(4)^2}{2}\right) + 5(3)\right]$ $= \frac{(n+2)^2(n+3)^2}{2} + 5n - 77$	
(iii) **[1]**	From (i) $\sum_{r=1}^{n} u_r = \frac{n^2(n+1)^2}{2} + 5n$ As $n \to \infty$, $\frac{n^2(n+1)^2}{2} \to \infty$, $5n \to \infty$ Hence, $\sum_{r=1}^{n} u_r = \frac{n^2(n+1)^2}{2} + 5n \to \infty$, series $\sum_{r=1}^{\infty} u_r$ does not converge.	

6.2 It is given that $\sum_{r=2}^{n} \frac{1}{r(r^2-1)} = \frac{1}{4} + \frac{1}{2(n+1)} - \frac{1}{2n}$.

(a) Show that $\sum_{r=2}^{n} \frac{1}{r(r^2-1)}$ is less than $\frac{1}{4}$. [2]

(b) Give a reason why the series $\sum_{r=2}^{\infty} \frac{1}{r(r^2-1)}$ converges, and write down its value. [2]

(d) Find $\sum_{r=m+1}^{N} \frac{1}{(r-m)(r-m+1)(r-m+2)}$, where m and N are integers with $N > m > 0$.
(There is no need to express your answer as a single algebraic fraction.) [2]

[VJC 2025 Prelim P1 Q6(a), (b), (d)]

Solution		Comments
(a) **[2]**	$\frac{1}{2(n+1)} - \frac{1}{2n} = \frac{n-(n+1)}{2n(n+1)} = \frac{-1}{2n(n+1)} < 0$ for all $n \geqslant 2$ $\sum_{r=2}^{n} \frac{1}{r(r^2-1)} = \frac{1}{4} + \frac{1}{2(n+1)} - \frac{1}{2n} < \frac{1}{4}$	
(b) **[2]**	As $n \to \infty$, $\frac{1}{2(n+1)} \to 0$ and $\frac{1}{2n} \to 0$, $\therefore \sum_{r=2}^{n} \frac{1}{r(r^2-1)} \to \frac{1}{4}$, a constant. Hence $\sum_{r=2}^{\infty} \frac{1}{r(r^2-1)}$ converges and $\sum_{r=2}^{\infty} \frac{1}{r(r^2-1)} = \frac{1}{4}$.	
(d) **[2]**	$\sum_{r=m+1}^{N} \frac{1}{(r-m)(r-m+1)(r-m+2)}$ $= \frac{1}{1(2)(3)} + \frac{1}{2(3)(4)} + \cdots + \frac{1}{(N-m)(N-m+1)(N-m+2)}$ $= \sum_{r=2}^{N-m+1} \frac{1}{(r-1)(r)(r+1)}$ $= \sum_{r=2}^{N-m+1} \frac{1}{r(r^2-1)}$ $= \frac{1}{4} + \frac{1}{2(N-m+2)} - \frac{1}{2(N-m+1)}$	

6.3 A sequence is such that $u_{n+1} = u_n + 1.5$, for $n = 1, 2, 3, \ldots$.

(a) Given that $u_{50} = 99u_1$, show that $u_1 = 0.75$. [2]

(b) Find the least value of n such that $u_2 + u_4 + u_6 + \cdots + u_{2n}$ is more than 2025. [2]

(c) It is given that u_{25}, u_k and u_5 are the first three terms of a geometric progression.

(i) Find k. [2]

(ii) Find the sum of the first 13 terms of the geometric series, giving your answer to 3 decimal places. [2]

[NYJC 2025 Prelim P1 Q8]

Solution		Comments
(a) **[2]**	Note that the sequence is an arithmetic progression with common difference 1.5 $u_{50} = u_1 + 49(1.5)$ $99u_1 = u_1 + 73.5$ $u_1 = 0.75$	
(b) **[2]**	**Method 1** $\sum_{k=1}^{n} u_{2k} = u_2 + u_4 + \cdots + u_{2n}$ is a sum of n terms of an arithmetic sequence with first term $u_2 = 0.75 + 1.5 = 2.25$ and common difference $2(1.5) = 3$ $u_2 + u_4 + \cdots + u_{2n} = \frac{n}{2}(2(2.25) + (n-1)(3))$ $= \frac{n}{2}(1.5 + 3n)$ From GC, for $\frac{n}{2}(1.5+3n) > 2025$, least $n = 37$ **Method 2** $\sum_{k=1}^{n} u_{2k} = \sum_{k=1}^{n} (2.25 + (2k-1)3) > 2025$ Using GC, least $n = 37$	
(c)(i) **[2]**	$\frac{u_k}{u_{25}} = \frac{u_5}{u_k}$ $u_k^2 = u_5 u_{25}$ $[0.75 + (k-1)(1.5)]^2 = [0.75 + (5-1)(1.5)]\,[0.75 + (25-1)(1.5)]$ $0.75 + (k-1)(1.5) = \sqrt{6.75 \times 36.75} = 15.75$ $k = 11$	
(c)(ii) **[2]**	**Method 1** Common ratio $= \frac{15.75}{36.75} = \frac{3}{7}$ $S_{13} = \frac{36.75\left(1 - \left(\frac{3}{7}\right)^{13}\right)}{1 - \frac{3}{7}}$ $= 64.31144$ $= 64.311$ **Method 2** Using GC, $\sum_{r=1}^{13} \left(36.75\left(\frac{3}{7}\right)^{r-1}\right) = 64.311$	

6.4 A sequence $u_1, u_2, u_3, \ldots$ is defined by

$$u_{n+1} = 2u_n\,(1-u_n)$$

for all positive integers n.

(a) Find the possible limits of the sequence, if the sequence is convergent. [2]

(b) For each of the following, evaluate u_2 and u_3, and state whether the sequence is convergent:

- $u_1 = 0$,
- $u_1 = 0.01$,
- $u_1 = -0.01$. [6]

[EJC 2025 Prelim P2 Q2]

Solution		Comments
(a) **[2]**	Let L be the limit of the sequence $u_{n+1} = 2u_n\,(1-u_n)$. $L = 2L(1-L)$ $2L^2 - L = 0$ $L(2L-1) = 0$ $L = 0$ or $L = \frac{1}{2}$	
(b) **[6]**	When $u_1 = 0$, $u_2 = 0$, $u_3 = 0$, sequence is convergent. When $u_1 = 0.01$, $u_2 = 0.0198$ (exact), $u_3 = 0.0388$ (3 s.f.), sequence is convergent. When $u_1 = -0.01$, $u_2 = -0.0202$ (exact), $u_3 = -0.0412$ (3 s.f.), sequence is not convergent.	

7 Arithmetic and Geometric Progressions

7.1 **(a)** **(i)** The ninth, fifth and second terms of an arithmetic progression are successive terms of a geometric progression with first term a and common ratio r, where $r \neq 1$ and $a > 0$. Find the value of r and deduce that the geometric series is convergent. [3]

(ii) Using the value of r found in (i), find the least value of n such that the sum of all the terms after the nth term of the geometric progression is less than 1% of its sum of the first n terms. [3]

(b) The sum, S_n, of the first n terms of another sequence is given by $S_n = n \ln 2^{q(n+1)}$, where q is a constant. Prove that the sequence follows an arithmetic progression. [4]

[ASRJC 2025 Prelim P1 Q8]

Solution		Comments
(a)(i) **[3]**	Let the first term and common difference of the arithmetic progression be b and d respectively. Since $b+8d$, $b+4d$ and $b+d$ are successive terms of a GP, then $\frac{b+4d}{b+8d}=\frac{b+d}{b+4d}$ (common ratio). $(b+4d)^2=(b+8d)(b+d)$ $8bd+16d^2=9bd+8d^2$ $d(8d-b)=0$ $b=8d \quad (d\neq 0)$ $\Rightarrow r=\frac{8d+4d}{8d+8d}=\frac{3}{4}$ Since $\lvert r\rvert=\left\lvert\frac{3}{4}\right\rvert<1$, the series is convergent.	
(a)(ii) **[3]**	$S_\infty - S_n < 0.01S_n \Rightarrow S_\infty < 1.01S_n$ $\frac{a}{1-\frac{3}{4}} < 1.01\left[\frac{a\left(1-\left(\frac{3}{4}\right)^n\right)}{1-\frac{3}{4}}\right]$ $1<1.01\left(1-\left(\frac{3}{4}\right)^n\right)$ $\left(\frac{3}{4}\right)^n<0.009901$ $n\ln\left(\frac{3}{4}\right)<\ln(0.009901)$ $n>\frac{\ln(0.009901)}{\ln\left(\frac{3}{4}\right)}=16.042$ Hence least n is 17.	
(b) **[4]**	$T_n=S_n-S_{n-1}=n\ln 2^{q(n+1)}-(n-1)\ln 2^{q(n)}$ $=n(n+1)q\ln 2-(n-1)(nq)\ln 2$ $=nq\ln 2+nq\ln 2$ $=2nq\ln 2$ $T_n-T_{n-1}=2nq\ln 2-2(n-1)q\ln 2$ $=2q\ln 2 \quad$ (constant $\because q$ is a constant) $\therefore$ the sequence is an arithmetic progression (Proved).	

7.2 An arithmetic series has first term a and common difference d, where a and d are non-zero real constants. A convergent geometric series has first term b and common ratio r, where $b>0$ and r is a non-zero real constant. The sum of the first n terms of the geometric series is denoted by S.

It is given that the 7th, 10th and 11th terms of the arithmetic series are equal to the 3rd, 6th and 10th terms of the geometric series respectively.

(a) Show that r satisfies the equation $3r^7 - 4r^3 + 1 = 0$. Solve this equation, giving your answer correct to 4 decimal places. [4]

(b) A new series is formed by taking the even numbered terms of the geometric series. Find the smallest value of n for which S differs from the sum to infinity of the new series by at least $1.78b$. [4]

[TMJC 2025 Prelim P1 Q4]

Solution		Comments
(a) **[4]**	$a+6d=br^2 \quad --(1)$ $a+9d=br^5 \quad --(2)$ $a+10d=br^9 \quad --(3)$ $(2)-(1): 3d=br^2(r^3-1) \quad --(4)$ $(3)-(2): d=br^5(r^4-1) \quad --(5)$ Substitute (5) into (4), $3br^5(r^4-1)=br^2(r^3-1)$ $3r^3(r^4-1)=(r^3-1) \quad$ (since $b, r\neq 0$) $3r^7-4r^3+1=0 \quad$ (shown) By GC, since the series is convergent, $\lvert r\rvert<1$, $r=0.6639$ (4 d.p.) [GC SCREEN PLACEHOLDER] GC 1: [APPS] PlySmlt2, POLY ROOT FINDER, ORDER 7 [NEXT] [GC SCREEN PLACEHOLDER] GC 2: coefficients $a_7=3,\ a_6=0,\ a_5=0,\ a_4=0,\ a_3=-4,\ a_2=0,\ a_1=0,\ a_0=1$ [SOLVE]	
(b) **[4]**	New Series: $br, br^3, br^5, \ldots$ $\left\lvert \frac{br}{1-r^2}-S\right\rvert \geqslant 1.78b$ $\left\lvert \frac{br}{1-r^2}-\frac{b}{1-r}(1-r^n)\right\rvert \geqslant 1.78b$ $\left\lvert \frac{0.6639b}{1-0.6639^2}-\frac{b}{1-0.6639}(1-0.6639^n)\right\rvert \geqslant 1.78b$ $\left\lvert \frac{0.6639}{1-0.6639^2}-\frac{1}{1-0.6639}(1-0.6639^n)\right\rvert \geqslant 1.78 \quad (\because b>0)$ Using GC, When $n=14$, $\left\lvert \frac{0.6639}{1-0.6639^2}-\frac{1}{1-0.6639}(1-0.6639^n)\right\rvert = 1.7785 < 1.78$ When $n=15$, $\left\lvert \frac{0.6639}{1-0.6639^2}-\frac{1}{1-0.6639}(1-0.6639^n)\right\rvert = 1.7818 \geqslant 1.78$ Therefore, least $n=15$.	

7.3 An arithmetic progression has first term a and common difference d, where a and d are non-zero. The 3rd, 9th and 11th terms of the arithmetic progression are three distinct consecutive terms of a geometric progression.

(a) Find d in terms of a. [3]

(b) Let the sum of the first n terms of the arithmetic progression be S_n where $S_1 = 143$. Find the largest possible value of S_n. [3]

The first term of the geometric progression is b, where b is positive.

(c) Determine whether the geometric series is convergent. [2]

(d) Show that the sum of all terms after the nth term of the geometric progression is at most $\frac{1}{2}b$, where n is a positive integer. [2]

[SAJC 2025 Prelim P1 Q5]

Solution		**Comments**
(a) **[3]**	$\frac{a+8d}{a+2d} = \frac{a+10d}{a+8d}$ $(a+8d)^2 = (a+2d)(a+10d)$ $a^2+16ad+64d^2 = a^2+12ad+20d^2$ $4ad+44d^2 = 0$ $4d(a+11d) = 0$ $d=0$ (reject since terms of AP, GP are distinct) or $a+11d=0$ $d = -\frac{1}{11}a$	
(b) **[3]**	$S_1 = a = 143$ $d = -\frac{1}{11}(143) = -13$ $S_n = \frac{n}{2}[2a+(n-1)d]$ $= \frac{n}{2}[286-13(n-1)]$ From GC, largest $S_n = 858$ [GC SCREEN PLACEHOLDER] GC 1: [Y=] Y1 = $\frac{X}{2}(286-13(X-1))$ [GC SCREEN PLACEHOLDER] GC 2: [2nd][TABLE] (TABLE SETUP: TblStart = 1, ΔTbl = 1), scroll down column Y1 to read the largest value	

(c) **[2]**	$r=\frac{a+8d}{a+2d}=\frac{-11d+8d}{-11d+2d}=\frac{-3d}{-9d}=\frac{1}{3}$ Since $\lvert r\rvert=\frac{1}{3}<1$, the geometric series is convergent.	
(d) **[2]**	Let u_n be the nth term of the geometric progression. $u_{n+1}+u_{n+2}+\ldots$ $=S_\infty-S_n$ $=\frac{b}{1-r}-\frac{b(1-r^n)}{1-r}$ $=\frac{br^n}{1-r}$ $=\frac{b\left(\frac{1}{3}\right)^n}{1-\frac{1}{3}}$ $=\frac{3}{2}\left(\frac{1}{3}\right)^n b$ $=\frac{b}{2}\left(\frac{1}{3}\right)^{n-1}\leqslant\frac{1}{2}b$ since $\left(\frac{1}{3}\right)^{n-1}\leqslant 1$ Alternative: $u_{n+1}+u_{n+2}+u_{n+3}+\ldots$ $=b\left(\frac{1}{3}\right)^n+b\left(\frac{1}{3}\right)^{n+1}+b\left(\frac{1}{3}\right)^{n+2}+\ldots$ $=\frac{b\left(\frac{1}{3}\right)^n}{1-\frac{1}{3}}$ $=\frac{3}{2}b\left(\frac{1}{3}\right)^n$ $=\frac{b}{2}\left(\frac{1}{3}\right)^{n-1}\leqslant\frac{b}{2}$ since $\left(\frac{1}{3}\right)^{n-1}\leqslant 1$	

7.4 Mabel and Janice decided to start a 5-year savings plan beginning in January 2026.

Mabel saves using a piggy bank. At the start of January 2026, she deposits \$101. Each subsequent month, she increases her deposit by \$1—so she deposits \$101 in January, \$102 in February, \$103 in March, and so on, until \$112 in December. At the start of each new year, she resets her monthly deposit to \$101 in January and repeats the same pattern through December. She continues this routine from 2026 to 2030, inclusive.

Janice, on the other hand, deposits \$100 at the start of every month into a bank account that earns 0.3% interest per month, with interest calculated and added into the account at the end of each month.

(a) Show that Janice will have more money in her savings account than Mabel has in her piggy bank at the end of December 2030. [5]

(b) Find the month and year when Janice's savings first exceed Mabel's savings. [4]

[TJC 2025 Prelim P1 Q7]

Solution | **Comments**

(a)
[5]

For Mabel:

Amount of money Mabel saves in a year

$=\$(101+102+\cdots+112)$

$=\$\frac{12}{2}(101+112)=\$1278$

Amount of money Mabel saves in 5 year $=\$1278\times 5=\$6390$

For Janice:

nth mth	Amount at start of month ($)	Amount at end of month ($)
1 Jan26	100	$100(1.003)$
2	$100+100(1.003)$ $=100(1+1.003)$	$100(1+1.003)(1.003)$ $=100\,(1.003+1.003^2)$
⋮		⋮
n		$100\,(1.003+1.003^2+\cdots+1.003^n)$ $=\frac{100(1.003)\,(1.003^n-1)}{1.003-1}$

At Dec 2030, $n=60$.

Therefore, Amount Janice saves in 5 years

$=\frac{100(1.003)\,(1.003^{60}-1)}{1.003-1}=\$6582.85$ (2 d.p.) [A1]

$\therefore$ Janice will have more money in her savings account than Mabel has in her piggy bank.

(b)
[4]

At end of 4th year,

Mabel will have \$1278 x 4 $=\$5112$

Janice will have $\frac{100(1.003)\,(1.003^{48}-1)}{1.003-1}=\$5169.97>\$5112$

At end of 3rd year,

Mabel will have \$1278 x 3 $=\$3834$

Janice will have $\frac{100(1.003)\,(1.003^{36}-1)}{1.003-1}=\$3806.97<\$3834$

Therefore, the year when Janice's saving is more than Mabel's saving happen in 4th year (i.e. 2029). Let it be the kth month.

Then set Janice's saving > Mabel's saving

$\Rightarrow \frac{100(1.003)\,(1.003^{36+k}-1)}{1.003-1}>3834+\frac{k}{2}\,[2(100)+(k-1)(1)]$

$\Rightarrow \left(\frac{100003}{3}\right)(1.003^{36+k}-1)-3834-\frac{k}{2}\,[199+k]>0$

Using GC, we have $k\geqslant 4$. Therefore, the month and year is Apr 2029.

7.5 Sarah, a Singaporean Citizen, started a 4-year undergraduate Computing course at the National University of Singapore (NUS) in August 2021. The annual tuition fee, after the MOE Tuition Grant, is \$8,250. Sarah took a Tuition Fee Loan (TFL) covering 90% of her tuition fees. This loan is interest-free during her course of study.

Sarah decided to set aside \$$x$ per month from August 2021. This amount is increased by \$2 each subsequent month. (Sarah set aside \$$x$ in August 2021, \$$(x+2)$ in September 2021, ….) With the amount set aside, she repaid one-fifth of her total TFL at the end of July 2025.

(a) Find the value of x. [3]

Upon Sarah's graduation, the TFL's interest rate becomes effective at a fixed rate of 0.4% per month, calculated on the outstanding balance at the end of the month. The interest is first charged on 31st August 2025.

Sarah plans to make monthly repayments of \$$y$ to the bank on the 1st of every month, starting from 1st September 2025.

(b) Find the amount of outstanding loan after Sarah's first monthly repayment. [1]

(c) Show that the amount of outstanding loan after Sarah's nth monthly repayment is

$$23760(1.004)^n - 250y\,(1.004^n - 1)\,.$$

[3]

(d) Sarah wants to pay off her remaining loan by the end of 3 years. Find the value of y. [2]

Sarah decided on a different repayment strategy. For the first 24 months, she repays \$500 per month. After these 24 months, she doubles her monthly repayment to \$1000.

(e) Determine the month and year Sarah will clear her loan and calculate the amount of her final repayment. [4]

[ACJC 2025 Prelim P1 Q12]

Solution		Comments
(a) **[3]**	Amount Sarah plans to repay (before graduation): $\frac{1}{5}\times\frac{90}{100}\times 8250\times 4=\$5940$ Number of repayments made from August 2021 to July 2025 is 48 (i.e. $n=48$). The repayments follow an arithmetic progression (AP) with first term $a=x$ and common difference $d=2$. The sum of an arithmetic progression is $S_n=\frac{n}{2}[2a+(n-1)d]$ $5940=\frac{48}{2}[2x+(48-1)2]$ $x=76.75$	
(b) **[1]**	Amount of outstanding loan after Sarah's graduation: $\frac{4}{5}\times\frac{90}{100}\times 8250\times 4=\$23760$ On 31st August 2025, apply interest of 0.4% to outstanding loan to get $\$(23760(1.004))$. On 1st September 2025, make 1st repayment of $\$y$ to the bank. $\therefore$ Outstanding loan after 1st repayment $=\$(23760(1.004)-y)$ or $\$(23855.04-y)$	
(c) **[3]**	On 30th September 2025, apply interest of 0.4%. On 1st October 2025, make 2nd repayment of $\$y$. Outstanding loan after 2nd repayment $=(23760(1.004)-y)\ 1.004-y$ $=23760(1.004)^2-1.004y-y$ Outstanding loan after 3rd repayment (on 1st Nov 2025) $=(23760(1.004)^2-1.004y-y)\ 1.004-y$ $=23760(1.004)^3-1.004^2y-1.004y-y$ Outstanding loan after n^{th} repayment $=23760(1.004)^n-1.004^{n-1}y-\cdots-1.004^2y-1.004y-y$ $=23760(1.004)^n-y\,(1+1.004+1.004^2+\cdots+1.004^{n-1})$ $=23760(1.004)^n-y\left(\frac{1.004^n-1}{1.004-1}\right)$ $=23760(1.004)^n-250y\,(1.004^n-1)$ (shown)	

(d) **[2]**	To repay remaining loan by the end of 3 years (i.e. 36 monthly repayments of \$$y$), let $n=36$. To fully repay the loan, we want outstanding loan amount after 36 repayments to be zero (or less). $23760(1.004)^{36}-250y\,(1.004^{36}-1)\leqslant 0$ $y\geqslant 709.9769$ $\therefore y=709.98$ (nearest cent)	
(e) **[4]**	After 24 monthly repayments of \$500 (i.e. $n=24$, $y=500$), outstanding loan amount $=23760(1.004)^{24}-250(500)\,((1.004)^{24}-1)$ $=13580.49$ Outstanding loan amount as of 1st August 2027 $=13580.49$ In a similar method to **(c)**, the outstanding loan amount after m monthly repayments of \$1000 (i.e. $y=1000$) is $13580.49(1.004)^{m}-250(1000)\,((1.004)^{m}-1)$ To fully repay the loan, we need $13580.49(1.004)^{m}-250(1000)\,((1.004)^{m}-1)\leqslant 0$ $m\geqslant 13.991185$ Least $m=14$ Sarah would require another 14 monthly repayments to fully repay her loan, with her first repayment of \$1000 on 1st September 2027. Hence, she would clear her loan on (1st) October 2028. The outstanding loan amount after her 13th monthly repayment of \$1000 (on 1st Sep 2028) is $13580.49(1.004)^{13}-250(1000)\,((1.004)^{13}-1)$ $=987.2529$ Interest is applied on 30th Sep 2028. Hence her final repayment amount is $987.2529\times 1.004=$ \$991.20 (nearest cent) (Alternatively, outstanding loan amount after 14th monthly repayment is $13580.49(1.004)^{14}-250(1000)\,((1.004)^{14}-1)$ $=-8.7981$ Hence the final repayment is $1000-8.7981=\$991.20$ (nearest cent).	

7.6 The dosage of medicine given to a patient needs to be carefully managed in order to achieve the desired result.

(a) To give the patient time to adapt to a particular medicine, the dosage is slowly increased over time. The patient is initially given a dosage of 0.75 units. The dosage is increased by 0.5 units each day until a total of at least 10 units of medicine has been taken. Find the least number of days it takes to achieve this, and the dosage on that day. [4]

(b) The doctor prescribed a new medication to another patient. After n days, the total dosage of the new medicine administered to the patient is given by

$$\frac{3}{20}\left(\frac{5^n - 4^n}{5^{n-1}}\right) \text{ units.}$$

(i) Show that the increase in dosage of the new medicine administered per day follows a geometric progression. [3]

(ii) Find the total dosage of the new medicine administered to the patient after a long period of treatment. [2]

(c) Doctors can manage the medication so that the daily dosage of medicine increases as an arithmetic progression. Explain in context why the arithmetic progression is not a preferred model for the dosage of medication. [1]

[RVHS 2025 Prelim P2 Q4]

Solution		Comments
(a) **[4]**	Let u_n be the dose on the nth day. $u_n = 0.75 + 0.5(n-1)$ $S_n = \frac{n}{2}(2(0.75) + (n-1)(0.5))$ $10 \leqslant \frac{n}{2}\left(1 + \frac{n}{2}\right)$ ——$(*)$ $\frac{1}{4}n^2 + \frac{1}{2}n \geqslant 10$ $n^2 + 2n \geqslant 40$ $(n+1)^2 \geqslant 41$ $n \geqslant -1 + \sqrt{41}$ or $n \leqslant -1 - \sqrt{41}$ (rej as no. of days cannot be -ve) $n \geqslant 5.40312$ Therefore, it takes 6 days. Alternatively, When $n = 5$, $\frac{1}{4}n^2 + \frac{1}{2}n = 8.75 < 10$ When $n = 6$, $\frac{1}{4}n^2 + \frac{1}{2}n = 12 \geqslant 10$ Therefore, it takes 6 days. $u_6 = 0.75 + 0.5(5) = 3.25$ The dose on the 6th day is 3.25 units.	

(b)(i) **[3]**	Increase on n^{th} day $= S_n - S_{n-1}$ $= \frac{3}{20}\left(\frac{5^n - 4^n}{5^{n-1}}\right) - \frac{3}{20}\left(\frac{5^{n-1} - 4^{n-1}}{5^{n-2}}\right)$ $= \frac{3}{20}\left(\frac{5^n - 4^n}{5^{n-1}} - \frac{5^{n-1} - 4^{n-1}}{5^{n-2}}\right)$ $= \frac{3}{20}\left(\frac{1}{5^{n-2}}\right)\left(5^{n-1} - \frac{4}{5}(4^{n-1}) - 5^{n-1} + 4^{n-1}\right)$ $= \frac{3}{20}\left(\frac{1}{5^{n-2}}\right)\left(\frac{1}{5}\right)(4^{n-1})$ $= \frac{3}{20}\left(\frac{4}{5}\right)^{n-1}$ which is of the form ar^{n-1}, where $a = \frac{3}{20}$ and $r = \frac{4}{5}$ which is a GP. **Alternative:** Consider $\frac{u_{n+1}}{u_n} = \frac{\frac{3}{20}\left(\frac{4}{5}\right)^{n+1-1}}{\frac{3}{20}\left(\frac{4}{5}\right)^{n-1}} = \frac{4}{5}$ which is a constant Since consecutive terms have a common ratio $r = \frac{4}{5}$, the increase in dosage follows a GP.	

(b)(ii)
[2]

Method 1

Long term dosage

$$= \lim_{n\to\infty} \frac{3}{20}\left(\frac{5^n - 4^n}{5^{n-1}}\right)$$

$$= \frac{3/20}{1-4/5}$$

$$= \frac{3}{4}$$

In the long term, the total dosage approaches 0.75 units.

Note:

After n days, the total dosage of the medicine is given by $\frac{3}{20}\left(\frac{5^n - 4^n}{5^{n-1}}\right)$.

Let S_n denote the total dosage of the new medicine after n days.

Then $S_n = \frac{3}{20}\left(\frac{5^n - 4^n}{5^{n-1}}\right)$.

Let v_n the change in dosage on nth day found in part (b)(i) where $v_n = S_n - S_{n-1}$

Since v_n is a geometric progression, we can use $S_\infty = \frac{a}{1-r}$, the sum to infinity formula to find the long term total dosage.

Method 2

$$\frac{3}{20}\left(\frac{5^n - 4^n}{5^{n-1}}\right)$$

$$= \frac{3}{20}(5)\left(\frac{5^n - 4^n}{5^n}\right)$$

$$= \frac{3}{20}(5)\left(1 - \left(\frac{4}{5}\right)^n\right)$$

As $n \to \infty$, $\left(\frac{4}{5}\right)^n \to 0$

Total dosage in the long term

$$= \frac{3}{20}(5)(1-0)$$

$$= \frac{3}{4}$$

(c)
[1]

If the dosage increases as an AP, then it will continue to **increase to** (positive) **infinity**.

8 Differentiation

8.1 A curve C has equation $x^2+y^2-4y=xy$, where $x \geqslant 0$.

(a) Show that $(2y-x-4)\dfrac{\mathrm{d}y}{\mathrm{d}x}=y-2x$. [2]

[RVHS 2025 Prelim P1 Q7(a)]

Solution		Comments
(a) **[2]**	$x^2+y^2-4y=xy$ $2x+2y\dfrac{\mathrm{d}y}{\mathrm{d}x}-4\dfrac{\mathrm{d}y}{\mathrm{d}x}=y+x\dfrac{\mathrm{d}y}{\mathrm{d}x}$ $(2y-x-4)\dfrac{\mathrm{d}y}{\mathrm{d}x}+2x-y=0$ $(2y-x-4)\dfrac{\mathrm{d}y}{\mathrm{d}x}=y-2x$	